

Poornachandra G.

Bioinformatics & Multi-Omics Leader | Precision Oncology | AI-Augmented Biomedical Data Science

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PROFESSIONAL SUMMARY

Bioinformatics and multi-omics leader with 10+ years in precision oncology, leading scientists and owning technical delivery across R&D, clinical, and biopharma collaborations, including Takeda, Genentech, Cleveland Clinic Florida and Bugworks. Translates complex biomedical data into decision-ready insights by integrating multi-omics with clinical, literature, and genetic evidence for target discovery and mechanism-of-action validation. Brings hands-on experience applying knowledge graphs to CRISPR-screen, drug-screen, and DepMap data, alongside LLM and RAG pipelines (LangChain, Azure AI Foundry) for literature mining and evidence integration. Combines technical roadmapping, team building, and cross-functional stakeholder leadership to deliver reproducible analytical solutions for precision-medicine and drug-discovery workflows.

SKILLS & TECHNOLOGIES

Multi-Omics & Biomedical Data	Multi-Omics Data Integration • Biomedical Data Science • WGS • WES • TES • RNA-seq • scRNA-seq • Spatial Transcriptomics • Liquid Biopsy • Clinical Genomics • Companion Diagnostics • Variant Interpretation (ACMG/AMP/ASCO/CAP) • MSI • HRD • TMB • CNV • SV • Karyotype • Bulk RNA-seq Deconvolution (CIBERSORTx, MuSiC) • Clinical-Trial Data Analysis • Biomarker Discovery • Therapy-Response Prediction
Clinico-omics Databases	Flatiron Health CGDB • ConcertAI datasets • ClinVar • ClinGen • OncoKB • CIViC • cBioPortal • GDC (Genomic Data Commons) • TCGA • CCLE • Cell Model Passports • LINCS • TARGET • HPA (Human Protein Atlas) • GEO (Gene Expression Omnibus) • Single-Cell Atlases • COSMIC
Target Discovery, Validation & Human Genetics	Computational Target Discovery • Target Validation • Target-Indication Assessment • Mechanism of Action (MOA) Validation • GWAS • Genetic Evidence Integration • Open Targets • DepMap • CRISPR-Screen Data Analysis • Drug-Screen Data Analysis • NetworkX • Neo4j • PyTorch Geometric (PyG) • Knowledge Graphs • Graph Analytics • Synthetic Lethality • Essentiality Scoring • Evidence-Linked Target Prioritization
Bioinformatics & Pipeline Engineering	NGS Pipeline Architecture • GATK • Sentieon • DRAGEN • NVIDIA Parabricks • FreeBayes • DeepVariant • DeepSomatic • PureCN • CNVKit • VEP • ANNOVAR • Nextflow • ETL Pipelines • Data Harmonization • QC Automation • Analytical Validation • Benchmarking • Reproducibility • Kallisto • Salmon • DESeq2 • edgeR • Seurat • Scanpy • GSEA
AI, Machine Learning & LLMs	Machine Learning • PyTorch • scikit-learn • LightGBM • XGBoost • Feature Engineering • SHAP • pandas • NumPy • SciPy • MLOps • MLflow • Large Language Models (LLMs) • LangChain • LangGraph • Retrieval-Augmented Generation (RAG) • RAG Pipeline Design • Azure AI Foundry • Azure AI Search • Prompt Engineering • Biomedical Literature Mining • Evidence Extraction • AI Evaluation Pipelines • Promptfoo • DeepEval • Golden Dataset Design • MCP
Programming & DevOps	Python • R • SQL • Bash • Docker • Git • CI/CD
Cloud & Data Platforms	AWS (EC2, S3, Batch, HealthOmics) • Azure • Databricks • DNAnexus • BaseSpace • Illumina

	Connected Analytics • Conda/Mamba • Jupyter • HPC
Scientific Delivery & Leadership	Technical Delivery Ownership • Technical Direction • Technical Roadmapping • BioPharma & External Partner Collaboration • Cross-Functional Collaboration • Team Building • Hiring • Mentoring • Project Prioritization • Resource Planning • Stakeholder Communication • Risk Management • Quality Management • HIPAA Compliance • Agile • Scrum • JIRA • Confluence

PROFESSIONAL EXPERIENCE

Cellworks - Lead Clinical Translation

2024 - Present

- Own technical delivery of multi-omics and clinical-genomics programs for precision oncology, translating NGS, variant, and clinical evidence into decision-ready insights for pharma, clinical, and internal R&D stakeholders.
- Served as technical and data advisor for a target-indication assessment with Bugworks, leading computational analysis of TCGA and cell-line omics profiles alongside drug-resistance perturbation data, including resistant-cell-line sequencing and gene-knockdown screens, to develop an evidence-linked candidate target list and validate compound mechanisms of action.
- Lead the bioinformatics team's technical roadmap across R&D and clinical workstreams; recruit and mentor scientists, set delivery priorities, and align resources against concurrent partner commitments.
- Built and deployed LLM-powered RAG pipelines using LangChain and Azure AI Foundry to mine biomedical literature, retrieve relevant evidence, and construct biological networks.
- Designed analytical validation studies for a multi-omics predictive-response program in NSCLC, establishing acceptance criteria, validation protocols, and quality checkpoints for immunotherapy decision support.
- Established risk-management and quality practices for bioinformatics pipelines, identifying data-quality failure points and implementing mitigation strategies before downstream delivery.
- Designed AI evaluation pipelines with Promptfoo and DeepEval, using golden datasets to assess retrieval accuracy and reduce hallucination in GenAI-assisted biomedical workflows.

Cellworks - Senior Scientist, Reporting Manager

2021 - 2024

- Built and led a bioinformatics team, owning recruitment, onboarding, mentoring, technical review, and day-to-day delivery of multi-omics interpretation work.
- Partnered directly with Takeda and Genentech stakeholders to deliver interpretation pipelines for clinical-trial datasets, translating genomic evidence into actionable therapy-response insights.
- Created input-agnostic workflows across FASTQ, BAM, and VCF formats and vendor data from 4BaseCare, Tempus, Caris, Foundation Medicine, and Alacris, aligning pipeline compatibility and data standards with external bioinformatics teams.
- Built cancer-driver and CNV-mRNA knowledge bases spanning 30+ cancer types from public multi-omics datasets, supporting evidence integration for target and biomarker assessment.
- Developed an Ensembl-based VUS reclassification pipeline that improved variant-classification accuracy by more than 40%, strengthening the genetic evidence available for clinical interpretation.
- Analyzed single-cell RNA-seq data from Human Protein Atlas normal-tissue samples with Seurat, Scanpy, and limma to establish in-silico control baselines for distinguishing cancer-specific expression signatures.
- Applied CIBERSORTx and MuSiC bulk RNA-seq deconvolution to resolve cancer-specific markers within heterogeneous tumor samples and improve multi-omics interpretation.

- Developed workflows for MSI, HRD, mutation signatures, TMB, and indel burden, connecting clinically relevant biomarkers to precision-oncology target and therapy decisions.

Cellworks - Senior Scientist, Clinical Translation

2018 - 2021

- Built ETL pipelines for DepMap multi-omics, CRISPR-screen, and drug-screen data, then applied knowledge-graph analytics and graph-based machine learning using NetworkX and PyTorch Geometric to identify essential genes and synthetic-lethal relationships.
- Led multi-omics analyses for MDS, AML, and MPN clinical studies to identify disease drivers, treatment-resistance and sensitivity biomarkers, progression signatures, and standard-of-care response predictors.
- Designed a custom liquid-biopsy panel and collaborated with Cleveland Clinic Florida stakeholders to develop an ML pipeline on longitudinal genomic data predicting early melanoma relapse.
- Developed multi-omics interpretation pipelines for 18,000+ Foundation Medicine cases, translating large-scale genomic data into structured clinical insights.
- Developed and benchmarked secondary-analysis pipelines using Illumina DRAGEN, NVIDIA Parabricks, and Sentieon, improving confidence in accuracy and turnaround-time performance.
- Compiled an inventory of NCCN-recommended and FDA-approved assays and tests spanning 88 cancer types to support clinical-genomics interpretation and downstream decision workflows.

Cellworks - Scientist, Clinical Translation

2017 - 2018

- Curated evidence and rationale libraries for therapy-prediction workflows, reducing turnaround time for live oncology case analyses by more than 40%.
- Analyzed 200+ oncology cases, applying cancer and drug-interaction networks to support target selection and therapy-response interpretation.
- Benchmarked sequencing pipelines across DNAnexus and MedGenome to establish reproducibility and confidence in clinical-genomics outputs.
- Developed variant annotation, classification and filtering framework to automate prioritization and standardize decision support according to ACMG guidelines.
- Acted as a technical bridge between engineering R&D and core development teams, accelerating product delivery through rapid prototyping and the automation of scalable data processing pipelines.

Cellworks - Associate Scientist

2015 - 2017

- Harmonized karyotyping, aCGH, IHC, and mutation outputs for 500+ cases through automated ETL and data-harmonization pipelines, creating a consistent foundation for multi-omics analysis.
- Standardized genomic-data interpretation with academic collaborators at Columbia University, University of Florida, Cedars-Sinai, Mayo Clinic, Washington University, and MMRF-Michigan.
- Automated report-generation workflows that reduced preparation time from approximately 15 hours to minutes per week, accelerating delivery of stakeholder-ready clinical reports.

INDEPENDENT PROJECTS

VarEffex: Built a multi-agent scientific literature-mining framework that extracts variant–function associations and links biomedical evidence to variant interpretation and computational target discovery.

PathoParse: Developed LLM-assisted and rule-based pipelines to convert unstructured pathology reports into structured clinical metadata, supporting downstream biomedical data analysis and evidence integration.

SQLChat: Created a multi-agent clinical data interface that translates natural-language questions into SQL, performs statistical checks, and returns interpretable summaries for research and stakeholder decision-making.

EDUCATION

B.Tech, Biotechnology
JNTUA CEP

2015

PUBLICATIONS & ABSTRACTS

Co-authored 9 peer-reviewed oncology publications and conference abstracts in J Neurooncol, J Clin Oncol, and Blood/ASCO, covering computational biomarker modeling, AML/MDS biosimulation, and therapy-response modeling.
Available on ORCID: [0009-0009-1610-1205](https://orcid.org/0009-0009-1610-1205)